

Step	Component	Mathematical Formulation	Interpretation	Comments
0	Problem Setup	$-\frac{1}{2}\psi_n''(x) + V(x)\psi_n(x) = E_n\psi_n(x), \quad x \in [-5, 5]$	The 1D time-independent Schrödinger equation.	The overall goal is to learn an unknown potential $V(x)$, wavefunctions $\psi_n(x)$, and eigenvalues E_n from noisy, low-fidelity data.
1	Synthetic Data	Spatial grid: $x \in [-5, 5]$ Noisy observations: $x \sim \mathcal{U}(-5, 5)$ $\tilde{\psi}_n(x_i) = \psi_n^{\text{true}}(x_i) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2)$	Synthetic data for physics enforcement.	Noisy observations simulate sparse experimental observations.
2	Neural Ansatz	$V_\theta(x) = \text{MLP}_V(\theta_V; x)$ $\psi_n^\theta(x) = \text{MLP}_\psi(\theta_\psi; x)$ $E_n^\theta = \text{learnable scalar}$	Learned potential and eigenstates	
3	Automatic Differentiation	$\frac{\partial}{\partial x} \psi_n^\theta$ $\frac{\partial^2}{\partial x^2} \psi_n^\theta$	Recover the first and second partial derivatives of the learned wavefunctions with respect to x .	
4	Total Loss	$\mathcal{L}_{\text{tot}} = \lambda_{\text{TISE}}\mathcal{L}_{\text{TISE}} + \lambda_{\text{norm}}\mathcal{L}_{\text{norm}} + \lambda_{\text{smooth}}\mathcal{L}_{\text{smooth}} + \lambda_{\text{data}}\mathcal{L}_{\text{data}}$	Low fidelity PINN objective function with four loss terms: Four loss terms: 1. Physics residual $\mathcal{L}_{\text{TISE}}$ 2. Wavefunction normalization $\mathcal{L}_{\text{norm}}$ 3. Potential regularizer $\mathcal{L}_{\text{smooth}}$ 4. Data mismatch $\mathcal{L}_{\text{data}}$	The weights assigned to each term are held constant throughout the training procedure.
5	Optimization	$\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}_{\text{tot}}$	Gradient-descent update	
6	Sanity Checks	See figure analysis in ./artifacts/figures.md.	Validate: • Orthogonality of eigenfunctions. • Energy ordering: $E_0^\theta < E_1^\theta < E_2^\theta$ • Smoothness of learned potential • Boundary decay behavior	Sanity checks via figure analysis is an essential component of the sanity check process and for providing interpretable insights.
7-9	Proper orthogonal decomposition (POD) Diagnostics	Construct snapshot matrix: $\Psi^\theta = [\psi_0^\theta, \dots, \psi_{N-1}^\theta]$ SVD: $\Psi^\theta = U \Sigma W^T$ where • the columns of U are POD modes • diagonal elements of Σ are singular values • I don't really care about W in this context, but what is the interpretation again?	POD of the learned wavefunctions.	The spectrum (whaat does this refer to again?) reveals learned structure where large gaps in consecutive singular values indicate low-rank structure is learned correctly.